Adult Income Classification: Supervised Modeling & Preprocessing (Day 2)

1. Preprocessing Choices & Pipeline Design

To ensure no data leakage and maintain a reproducible workflow, we built a Scikit-Learn `ColumnTransformer` to handle our mixed dataset strictly on the training folds.

- **Numeric Pipeline:** We used `SimpleImputer(strategy='median')` followed by `StandardScaler`. Median imputation was deliberately chosen over the mean because features like `capital-gain` are highly skewed; the median is robust to these extreme outliers. The `StandardScaler` ensures all numeric features have zero mean and unit variance, which is critical for distance-based or regularized models like Logistic Regression to converge efficiently. (We skipped `KNNImputer` to prioritize training speed).
- **Categorical Pipeline:** We used `SimpleImputer(strategy='most_frequent')` followed by `OneHotEncoder(handle_unknown='ignore')`. `OneHotEncoder` was selected because features like `occupation` and `native-country` are nominal without an inherent ranking. Forcing an `OrdinalEncoder` on these would confuse linear algorithms. Setting `handle_unknown='ignore'` acts as a safety net against rare categories appearing in the test set.

2. Model Metrics & Evaluation

We trained two supervised models—Logistic Regression (with L2 regularization via the `lbfgs` solver) and a standard Decision Tree—and evaluated them against our Day 1 baseline on the 20% hold-out test set.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC	PR AUC
Day 1 Baseline (Rule)	75.30%	48.44%	49.70%	0.491	0.665	0.361
Decision Tree	82.16%	63.38%	59.67%	0.615	0.825	0.638
Logistic Regression	**85.24%**	**73.75%**	59.37%	**0.658**	**0.903**	**0.758**

Interpretation: Both machine learning models vastly outperformed the Day 1 baseline. The Logistic Regression model achieved an exceptional Precision of **73.75%**, meaning nearly 3 out of 4 individuals it predicts as high-earners are truly high-earners. This perfectly aligns with our business goal of minimizing wasted outreach. While False Negatives (missed high-earners) remain higher than False Positives across both models, this is an acceptable trade-off for protecting the marketing budget.

3. Interpretability Check

- **Logistic Regression:** Mapping the coefficients back to the One-Hot Encoded features revealed sensible logic. Wealth (`capital-gain`), being married (`Married-civ-spouse`), and holding management roles (`Exec-managerial`) strongly pushed the model toward predicting `>50K`. Conversely, working in `Priv-house-serv` or being `Never-married` strongly pushed predictions toward `<=50K`.

- **Decision Tree:** The raw decision tree suffered from severe overfitting, reaching a depth of 52 and scoring 97.2% on the training set but dropping to 82.1% on the test set. However, its top three splits (`marital-status`, `capital-gain`, and `education-num`) completely agreed with the top drivers identified by the Logistic Regression coefficients.

4. Model Selection & Next Steps

Going into Day 3, we will continue developing **both** model families, but with different ultimate goals:

1. **Logistic Regression:** Because it scales perfectly, avoids overfitting, and offers high interpretability, it will serve as our primary production candidate if we require a fast, highly-precise linear model.
2. **Tree-Based Models:** Despite the raw Decision Tree's overfitting flaw, tree algorithms naturally handle non-linear relationships much better than linear models. Therefore, we will upgrade the Decision Tree into a **Random Forest** or **Gradient Boosting Classifier (XGBoost)**. By using an ensemble approach to control the overfitting depth, we expect the tree-based models to ultimately surpass the Logistic Regression model in our primary F1-Score metric.

Preprocessing changes to test tomorrow: Log-transforming the highly skewed `capital-gain` feature, and binning the high-cardinality `native-country` feature to reduce noise.